

these boards were PCIe-based, and all the boards featured SATA support—and most of them had the standard four SATA connectors found on the higher end boards of six months ago!

While buying a motherboard, you must keep in mind its form factor (size). Boards primarily come in two sizes, ATX (the regular big boards requiring a nice big cabinet!) and Micro ATX (tiny boards needing a relatively smaller cabinet, and hence ideal for office environments or places where desk real estate is of prime importance).

The Micro ATX form factor has its own issues during assembly, as a PC built on this form factor will, in all probability, have a small cabinet. A small cabinet implies fewer avenues for proper ventilation and hence heating issues.

This is not to say that Micro ATX is useless. We are just trying to emphasise that while building a system of this form factor, you must be attentive to system ventilation.

You also need to keep in mind socket support. As of now, the most common AMD sockets are the 754 (the Athlon 64 range) and the 939 (the Athlon 64 as well as the Athlon 64 FX range).

939 is the latest socket for the Athlon range (the Opteron runs on 940 and is for servers anyway)

and it is advisable that you go for this socket over the 754, as it will give you a fair bit of future-proofing.

Coming back to the feature set, all the boards we received supported at least 2 GB of DDR 400 RAM, a feat that was commendable for high-end boards a year ago! In fact, the MSI RS480M2-IL and the ASRock 939A8X-M went a few yards further and offered support for a whopping 4 GB of RAM, plenty for all you might want to do for at least the next two years if not more (we wouldn't want to do a Bill Gates by predicting how much memory you might have use for!)



TUL A480M7-V

We had three boards based on the spanking new nForce 4 chipset from nVidia—these were the Winfast NF4K8AB, NF4K8MC and the Gigabyte GA-K8NE, and all these supported 3 GB of RAM—a peculiar number if you ask us.

Onboard sound has improved by leaps and bounds from the 2.1 systems we used to get a couple of years ago

A lot of the boards we received did not have onboard graphics, which is rather strange as these boards are targeted at an audience looking at complete on-board solutions.

All the nForce4 boards, along with the ATi RS 480-based board from MSI (the MSI RS480 M) lacked onboard video, but did feature an x16 PCIe slot, and were hence tested with a PCIe-based nVidia 6200 graphics card. Their AGP counterparts weren't any better, and we received no less than six motherboards without onboard video but featuring an AGP slot. These were tested with an nVidia 6200 AGP card.

Here, we must mention that only the Krypton M7VIG400 featured an extremely dated AGP 4x, whereas all the others had at least AGP 8x.

There were boards which were extremely generous in terms of their video solutions: both the TUL boards came with onboard video as well as an x16 PCIe slot. This feature was also available on the MSI RS480M2-IL.

Onboard sound has improved by leaps and bounds from the 2.1 systems we used to get a couple of years ago. Half the boards we tested featured eight-channel (7.1) audio support, and the other half featured six-channel (5.1) audio onboard, so basically, six-channel has become pretty much a stan-

How We Tested

In order to judge which motherboard was best, they were passed through a series of rigorous benchmarks. As many as five test-bench setups had to be used due to the various types of processor sockets on the motherboards that were tested.

Intel motherboard setups consisted of the Pentium IV 3.6 GHz processor for the LGA 775 socket, and the Pentium IV 3.2 GHz processor for Socket 478 boards. AMD motherboard setups consisted of the Athlon64 FX-53 for Socket 939, Athlon64 3000+ for Socket 754 and Athlon XP 2500+ for Socket 462 boards.

Other than that, the test setups consisted of 512 MB of DDR 400 MHz RAM as the need arose, a 250 GB 7200 rpm SATA Maxtor hard drive, and an nVidia GeForce 6200 PCIe or AGP display card where onboard display was not available. Windows XP Pro with SP1 was freshly loaded along with DirectX 9.0c, and the drivers from the driver CDs provided with the motherboards were loaded.

The various features of the motherboards were noted and rated accordingly. The tests and benchmarks consisted of the following:

For the game tests, we used relatively lower resolutions—640 x 480 and 800 x 600—as the onboard gaming/graphics solutions on these boards, though quite good, were not what we would term as powerful.

Serious Sam Second Encounter: This is a Direct3D game, and it accurately measures the Direct3D gaming subsystem. This game

was chosen because it runs fairly well on systems that use onboard graphics.

Quake 3 Arena: This OpenGL game which is fairly old was used to test the strength of the OpenGL graphics subsystem. This game was chosen because it scales fairly well across various systems.

3DMark 2001 SE: This is an industry-standard Direct3D synthetic benchmark and measures the graphics performance and capabilities of a system. This test gives one an indication about the kind of graphics performance you can expect from the system.

PC Mark 2004: This benchmark from FutureMark measures the CPU, memory, graphics and hard disk performance of the system, and hence is a good system-wide benchmark.

SiSoft Sandra 2005: SiSoft Sandra 2005 consists of benchmark modules that measure CPU, file system and memory performance.

Ziff-Davis Business Winstone 2004: This benchmark suite runs various applications such as WinZip, MSOffice 2002's Word, Excel, PowerPoint and Outlook, and other software, to measure the real-world performance of the system.

Dr. DivX 1.06: This is a video-encoding application that is used to encode DivX video files. Since video encoding is a processor-intensive task, the time taken to encode a file pretty much reflects how fast the processor is. A 100 MB DVD file was encoded to DivX, and the time taken was noted.